

Pop Quiz 1

*Total Marks: 5**06 August 2026*

1. Let A_1, A_2 be DFA with k_1, k_2 states respectively. Let $L = L(A_1) \cup L(A_2)$. Suppose $L \neq \emptyset$ **and L is infinite**. Then which of the following statements must be true. [4×0.5 marks]
- (a) There exists $w \in L$ such that $|w| < \max(k_1, k_2)$. **True**
 - (b) There exists $w \in L$ such that $|w| < k_1$. **False**
 - (c) For every $w \in L$, $|w| > \max(k_1, k_2)$. **False**
 - (d) There exists $w \in L$, $\max(k_1, k_2) < |w|$. **True**

2. Show that the language of all words which do not contain aba as a subword is regular by drawing a DFA for it. [**2 marks**]

Answer: For (a), since $L \neq \emptyset$, there exists some $w \in L$.

- If $L(A_1) = \emptyset$ (or symmetrically $L(A_2) = \emptyset$; note that both can't be empty), then $L = L(A_2)$, and there is an accepting path of length at most $k_2 - 1$, i.e., we find $|w| < k_2 \leq \max(k_1, k_2)$.
- If both $L(A_1), L(A_2)$ are non-empty, then let u be shortest word accepted by A_1 and v be shortest word by A_2 . If $|u| < |v|$ (symmetric argument for $|u| > |v|$), $u \in L$ and $|u| < |v| \leq k_2 - 1 < \max(k_1, k_2)$, again using the fact that in A_2 there is an accepting path of length at most $k_2 - 1$. Only case left is if $|u| = |v|$. But in this case, $u \in L$ and $|u| \leq k_1 - 1 \leq \max(k_1, k_2)$.

For (b), answer is clearly not true as it only depends on k_1 . For (c), it cannot hold either as its the negation of (a). Finally, for (d), it is true since language L is infinite.

[**1 mark**] for attendance